Pre-registration (DRAFT) — An Anomaly-Residual Search for Unexplained Transit and Photometric Signatures Around Main-Sequence K Dwarfs

Status: DRAFT / in preparation — NOT yet registered. Phase 2 of the program whose Phase 1 (white dwarfs) is registered at OSF 10.17605/OSF.IO/6YH7R. This plan is to be frozen and registered on OSF **before any K-dwarf light curve is analysed**; until then it is edited freely in this repository. It reuses the *validated* population-agnostic pipeline core (see `pipeline/core/`, regression-tested in `pipeline/runners/validate_wd.py`); the K-dwarf-specific parts are a new population plugin, not a fork.

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0. Summary

A pre-registered, mechanism-agnostic search for anomalous transit and photometric signatures around **main-sequence K dwarfs** — the longest-lived and most thermally-stable *living* stellar hosts — restricted only by an *activity-based youth floor* (§3) that excludes the genuinely young. Phase 1 applied this anomaly-residual methodology to white-dwarf remnants and returned a clean, fully-explained null with a quantitative upper limit. Phase 2 carries the **same validated machinery** to a living host, with **transit morphology as the primary channel**. We search for departures from the natural model — most powerfully, transit light curves whose *shape* no natural occulter (planet, eclipsing binary, disintegrating body, starspot) can produce — and try to explain each away through a fixed natural-explanation battery, with detection thresholds set by an empirical null and injection-recovery, never by inspecting candidates.

1. The search target and its assumptions

- **The target is the observable anomaly** — an unexplained transit/photometric departure — not a presupposed structure, mechanism, or agent (as in Phase 1).
- **The agnostic stance, extended.** We are agnostic about: (a) *mechanism* — no assumed engineering; (b) *engineering vs. mere existence* — we detect departures in the star's light and do not presuppose they are built; (c) *timescale* — we assume no schedule of activity. A consequence we state plainly: over our ~decade observational baselines, any process on

  > decade timescales is effectively **static**, so the realistic primary signal is a *steady* anomaly (a persistent anomalous transit shape), with genuine *variability* (a changing transit depth or morphology) the highest-value-but-lowest-prior bonus, not the expectation.

- **The one prior we retain: time for life to have *originated*.** This is not a mechanism or timescale assumption — it is the precondition for there being anything to find at all. K dwarfs uniquely maximise it: 15–40 Gyr main-sequence lifetimes mean every K dwarf has had ample time and none has evolved off the main sequence. The sample is further weighted toward *old* K dwarfs (§3).

- **What we refuse to assume.** We do not assume life requires planets, photosynthesis, visible light, or any particular biochemistry. None of these enters the detection — which measures only departures in the star's integrated light — so the search takes no position on them.
- **Why K dwarfs, and the honest limitation.** Among *living* hosts, K dwarfs offer the longest stable lifetime, the quietest baseline (calmer than flaring M dwarfs), and — being small — the deepest, most morphologically-resolvable transits. The cost, stated up front: a living star's cool photosphere is bright in the infrared, so the IR-excess channel that anchored Phase 1 loses its contrast. **Phase 2 is therefore transit-anchored, not excess-anchored.** We further acknowledge the transit channel is comparatively well-trodden; our differentiator is not the channel but the *discipline* — calibrated empirical-null thresholds and a mandatory, pre-registered explain-away battery applied uniformly.

2. Research questions

- **RQ1 (morphology).** Do any K dwarfs show a periodic transit whose *shape* — asymmetry, flat-bottomedness, box-versus-U, anomalous duration or secondary structure — cannot be reproduced by any natural occulter (sphere, ring, eclipsing binary, disintegrating body) after those are explicitly fitted and excluded?
- **RQ2 (photometric departure).** Do any show a persistent or time-varying photometric signature departing from the natural rotation/granulation/activity model that survives the battery?
- **RQ3 (upper limit).** With no unexplained anomaly, what upper limit `f_max` can we place on the prevalence of anomalous occulters/departures around main-sequence K dwarfs (above the youth floor), as a function of depth and period?

3. Sample and inclusion

- **Parent sample.** Main-sequence K dwarfs from Gaia DR3: Teff 3900–5300 K, log g > 4.3 (dwarf, excluding subgiants/giants), with clean astrometry (RUWE < 1.4, parallax_over_error > 10), plus the equivalent absolute-G versus (G_BP − G_RP) main-sequence box. *Scale (this repository, sized pre-data):* ~12,600 at G < 11; ~22,000 within 100 pc; ~178,000 at G < 13. Target finding is not the binding constraint. The exact constraints (the colour–magnitude box, the RUWE and parallax cuts, and the hard magnitude/volume limit) are committed as an **immutable, checksummed manifest** of source_ids before any light curve is pulled — the frozen sample against which all later results are reproducible, identical in spirit to the Phase-1 `wd_sample` manifest.
- **Data-quality / channel requirement.** A usable TESS and/or Kepler light curve (the transit instrument). The Kepler-field subset (4-year continuous photometry) is the morphology gold standard and is treated as a distinct, deepest tier.
- **The goal is the whole population; staging is a resource constraint, not a scientific one.** We impose no scientific exclusion beyond the minimal *youth floor* below and the data-quality requirement above. Which stars have been *analysed* at

any moment is set by available compute; the asymptotic target is the entire K-dwarf census.

- **Youth floor (the one real cut — an *activity-based youth proxy*).** Its purpose is the *life prior*: a star must have *existed* long enough for life to have had the chance to originate. But we are honest about the observable: **we cannot measure a field K dwarf's age directly** (no per-star asteroseismology, no cluster membership), so the floor is operationally an **activity/rotation cut interpreted as youth** through the well-established age–rotation–activity relation — a fast rotator / X-ray-saturated / highly variable K dwarf is young. The intent is age; the measurement is activity; we state both rather than dress one as the other. The floor is a **fixed rule, not a placeholder** (so the data decides *which* stars, no human break-picking): a star is excluded iff **(i)** its gyrochronological age from a measured rotation period, via the Angus et al. (2019) age–rotation relation, is **< 1.0 Gyr**, **or (ii)** its fractional X-ray luminosity (eROSITA/ROSAT) is **$\log(L_X/L_{bol}) > -4.0$** (saturated/active), **or (iii)** it lacks a rotation period *and* sits in the youngest decile of the Gaia photometric-variability-amplitude distribution. Stars with no measurable rotation period and normal activity are **retained** (absence of a period is not youth). These relations and the three numeric cuts (1.0 Gyr, −4.0 dex, 10th percentile) are frozen here; rotation/activity are measured pre-unblinding as noise characterisation, never from the transit candidates.
- *Reconciling with "no quiet cut" below.* The youth floor and the no-quiet-cut principle sit at **opposite ends of the same activity axis, for different reasons**. The youth floor removes only the *extreme* young/active tail — and removes it for the **life prior**, not for noise. Everything above it (the broad spread of moderately-active-to-quiet *old* stars) is **retained**, and its residual activity is handled by per-star self-weighting, **not** by a further activity cut. So we do cut on activity once, at the young extreme, to serve the age prior; we do *not* cut on activity again to manage noise.
- **No hard "quiet" cut — noisy stars self-weight.** Rather than excluding active stars, we include every star above the youth floor and let the per-star injection-recovery completeness C_i (§5) do the work: a transit is harder to recover in an active light curve, so a noisy star earns a low C_i and contributes little to `f_max` automatically (the Phase-1 self-weighting principle — under-probed objects weight toward zero). The limit stays honest over whatever has been analysed.
- **Resource-triggered, nested, staged analysis order (so iteration stays confirmatory).** Because the goal is everything and the binding constraint is compute, we pre-register a *nested* sequence of tiers (cleanest/brightest first: $T_1 \subset T_2 \subset \ldots \subset$ the full sample), analysed in that fixed order, with the expansion trigger being *available resources*, declared in advance — never the result of a prior tier. **The detection threshold is *not* spent per stage** (our compute-determined tiers have no pre-fixed information fractions, so a group-sequential alpha-spending function would not apply): instead the survey-wide family-wise threshold (§5) is set **once, for control over the entire intended census** — the full frozen sample size N_{total} declared in the manifest — and applied **unchanged** to every tier. Expanding the analysed set can therefore only add confirmed detections under the *same fixed bar*; it can never relax the threshold, so there is no dial to tune by stopping at a convenient tier. This is deliberately conservative (we control as if the whole census

were analysed from the start). Growing the sample is a pre-registered confirmatory extension (cf. the Phase-1 variability expansion); what is forbidden is loosening any boundary *in reaction to a result* or to rescue a candidate.

- No assumption that an anomaly orbits in a habitable zone; all periods the data permit are searched.
- **Stated selection biases:** brightness-limited (toward nearby), the TESS/Kepler footprint, and the youth floor. `f_max` is interpreted as a prevalence among the *effectively probed, solar-neighbourhood* K-dwarf population, self-weighted by per-star completeness.

4. Channels

- **Channel B — transit morphology (PRIMARY, CALIBRATED).** Detection by BLS/TLS for periodic box/U transits, **plus a parallel variable-depth / aperiodic-dip detector**, then morphology metrics (asymmetry, flat-bottom fraction, box-vs-U, duty cycle) and a **mandatory automated difference-image centroid gate** (§5). **Detection does not depend on morphology.** BLS/TLS (and the variable-depth detector) are the gate: any sufficiently deep dip is *caught as a candidate* regardless of its shape — a smooth, U-shaped occulter (e.g. an end-on cylinder) is detected exactly as a planet is. The morphology and depth-variability metrics are *parallel safety nets* that run **after** detection to flag the non-planetary minority; they are never required to trigger a detection. This is deliberate: when the baseline is a rigorous null, an unexplained anomaly is already the extreme tail, and demanding it *also* cast a diagnostically asymmetric shadow would be asking for the tail of the tail. We do not — detection sensitivity (the `f_max` in §5) rests on the depth/period reach of BLS, not on resolving exotic morphology. The Phase-1 machinery transfers directly: K dwarfs are point sources with clean Gaia astrometry, and a real occulter gives a *deep* transit, so the “deep ⇒ on-target real; shallow-and-offset ⇒ background blend, confirmed by centroiding” logic applies (the same code validated for white dwarfs, `pipeline/analysis/07–09`). Crucially, the sample size — tens of thousands versus Phase 1’s 157 — **graduates this from a secondary, candidate-generating channel into a primary, calibrated one**: it now supports a rigorous population-level upper limit on structural transit anomalies (RQ3, §5), the transit analogue of Phase 1’s IR-excess `f_max`.
- **Variable-depth detection is not optional.** BLS/TLS are optimised for *constant-depth* periodic boxes, but the highest-value signals (§1) — and a natural battery item, the disintegrating dust tail (KIC 12557548) — have transit depths that *fluctuate epoch-to-epoch*. A BLS-only trigger would down-weight or miss exactly the dynamic morphologies we most want to evaluate, blinding the search to its own stated priority. Detection therefore runs the variable-depth/aperiodic pass (single-event and depth-varying matched filters; cf. KIC 8462852) *alongside* BLS/TLS, with its own empirical-null calibration (§5).
- **Morphology resolution is cadence-bounded (stated caveat).** Resolving Arnold-type ingress/egress asymmetry requires fine sampling: at TESS 2-minute cadence a ~2-hour transit’s ingress is only a handful of points, so the subtlest non-spherical signatures are resolvable only for the brightest targets with 20-second cadence,

long-duration transits, or deep events. The morphology channel's sensitivity is reported *as a function of cadence and transit duration*, not assumed uniform.

- **Channel P — photometric departure (SECONDARY).** Departures from the natural rotation/granulation/activity model; correlated/structured variability; same empirical-null calibration as Phase 1.
- **Channel A — infrared excess (CALIBRATED but weak; self-weighting + corroborating).** A K dwarf's cool IR photosphere suppresses excess contrast and debris disks are a common, *expected* natural explanation, so this channel is *weak* — but weak is not zero, and **we keep a calibrated `f_max`** (correcting an earlier draft that dropped it). The Phase-1 self-weighting math already handles the weakness exactly: inject a synthetic anomalous excess and measure per-star completeness C_i; where the photosphere swamps the signal (a cold, low-`f` anomaly) injection-recovery returns C_i ≈ 0 and that star contributes nothing to `f_max = 3/ΣC_i`, while a warm, luminous structure (e.g. 500 K at `f`=0.1) gives C_i ≈ 1 and a real constraint. The limit therefore *self-evaporates where the data are blind and stands where they are not* — dropping it pre-emptively would throw away a valid (if weak) upper limit on bright technological structures. We state plainly that this limit is inherently weaker than Phase 1's WD limit, and let ΣC_i show by how much. Natural debris disks are removed by the battery (SED regime + SIMBAD/literature cross-check), exactly as for white dwarfs.
- **Corroborating extreme-outlier flag — with a *locked* threshold (coincidence multiplicity).** Separately from the `f_max`, Channel A contributes a high-bar flag for the *far tail* (an excess outside any natural regime, an anomalous SED shape, or a *fluctuating* excess a static disk cannot fake). Because we elevate a star when its IR flag **coincides** with a Channel-B transit anomaly, the intersection has its own multiplicity: with M Channel-B survivors and K Channel-A flags over N stars, the expected chance coincidences under the null are ≈ M·K/N. K must therefore be *locked before unblinding*, not chosen to manufacture a coincidence: the Channel-A flag fires at a fixed **> 4σ on the per-cohort IR empirical null**, which fixes K, so the expected false-coincidence rate M·K/N is pre-computed and registered here (with M held to ≈1 by the §5 FWER bar, a 4σ tail giving the joint chance rate ⩽ 1). A surviving extreme IR excess *and* an anomalous transit on one K dwarf is then the high-value residual — far stronger than either channel alone, and now with a pre-registered false-alarm budget rather than an arbitrary one.

5. Natural-explanation battery, statistics, and stopping rule

A flagged candidate becomes a residual only if it survives every applicable test. The order is deliberate: the cheapest, highest-yield contaminant filter runs *first and automatically*, because at this sample size nothing can be vetted by hand.

1. **Background eclipsing binary / aperture blend — automated centroid gate (FIRST, mandatory).** TESS's ~21″ pixels guarantee that *hundreds* of off-target eclipsing binaries blend into K-dwarf apertures — the dominant transit false positive at scale. Difference-image centroiding (the flux-weighted centroid of the out-of-transit minus in-transit image) is run **automatically as the first gate**: a candidate whose dip centroid sits more than a pre-set tolerance (≈1 TESS pixel) from the

target is killed with no human inspection. In Phase 1 this was a manual check on 3 candidates; here it is a mandatory, automated guillotine at the front of the battery.

2. **Eclipsing binary (on-target)** — fit EB models; check for a secondary eclipse, odd–even depth differences, ellipsoidal/reflection modulation; radial velocities where available.
3. **Stellar activity** — starspot rotational modulation, flares, faculae; cross-check against the star's rotation period and activity indicators.
4. **Bona-fide planet** — a real planet transit is *natural*; we flag only morphologies inconsistent with a spherical or ringed planet (noting the ringed-planet ambiguity of Arnold 2005).
5. **Disintegrating planetesimal / dust tail** — tested by recurrence, depth evolution, and an explicit asymmetric dust-tail template (cf. WD 1145+017, KIC 12557548).
6. **Instrumental / systematic** — TESS/Kepler systematics, momentum dumps, scattered light, aperture contamination.

- **Detection threshold and the trial factor.** With tens of thousands of stars × $\sim 10^6$ BLS period/phase trials each, a 3–4σ bar would bury the residual list in noise. We therefore compose two corrections, both pre-specified as *procedures* (the resulting numbers come from the data via those procedures, never from inspecting candidates):
- **(a) Per-cohort empirical null — with an outlier-*blind* noise metric.** An empirical null on the per-star BLS detection statistic, computed *within noise/activity cohorts* (binned by each light curve's scatter / activity level) rather than globally — so a quiet star is judged against other quiet stars and the active stars' heavy-tailed noise (spots, flares, rotation) does **not** inflate the threshold for the clean ones. **The cohort-assignment noise metric is an outlier-blind robust estimator — the median absolute deviation (MAD) of the out-of-transit continuum after iterative σ-clipping of *downward* excursions (points > 3σ below the median are masked) — never the raw standard deviation or CDPP.** This is load-bearing: a genuine deep anomalous dip inflates a naïve variance, which would mis-bin that very star into a "noisy" cohort, raise its local threshold, and let the anomaly **mask itself**. Clipping the dips before measuring the baseline closes that data-leak. Within each cohort, genomic-control inflation λ rescales for the non-Gaussian reality of TESS/Kepler systematics (as the WISE-excess null did, λ≈10.6 in Phase 1); this per-cohort grading (the v2 W2-offset lesson, generalised) is what lets us keep every star above the youth floor without a hard activity cut.
- **(b) One survey-wide family-wise (FWER) bar — a single framework, not a mix.** The threshold controls the expected number of pure-noise false alarms across the *entire frozen manifest* to **< 1** (cf. the Kepler 7.1σ bar, Jenkins et al. 2002, for ~150k stars), derived from **our own injection-recovery**, not borrowed — it depends on our sample size, cadence, and period range, and is expected to land in the **≈6–7σ regime**. Crucially we **do not compose mismatched error-control frameworks**: we control FWER, full stop; we do *not* also spend FDR against the same decision (FDR / Benjamini–Hochberg is used only to *rank-report* the surviving residual list, never to set the detection bar), and we do *not* spend alpha per tier — `N_total` is the fixed size of the immutable manifest (§3), not an open-ended census, so the bar is well-defined and set **once** for the whole population.

- **(c) Calibrated up front from the whole manifest, then frozen.** Both the per-cohort nulls and the FWER bar are calibrated **once, from the noise floor of the *entire* manifest** — a cheap pass computing each star's robust scatter and BLS-null statistic, with **no transit search** — *before* any tier is searched. This is what makes the staged analysis (§3) statistically clean: because the cohorts and the bar are fixed from the whole population at the outset, analysing tiers in compute order cannot retrospectively shift the null or threshold a later tier sees (the "cohort clash" is closed). Only the transit search and injection-recovery are staged; the statistical calibration covers everything from the start. The bar is frozen before any real candidate tail is unblinded.
- **Upper limit (Channel B, now calibrated) — reported *per morphology family.*** The Poisson zero-detection bound `f_max = 3 / Σ_i C_i`, with per-star completeness C_i measured by **injection-recovery of a frozen library of forward-modelled anomaly morphologies into the real light curves**. The library is registered here (not chosen after seeing data) and generated by a single exact procedure — sweeping an opaque/graded occulter mask across a limb-darkened stellar disk (`core/transit.py`): a **flat occulter** (box), an **asymmetric tilted / triangular occulter** (Arnold-type), and a **disintegrating dust-tail** (sharp ingress / slow egress with per-epoch depth variation and dropouts), all against a **natural limb-darkened-planet negative control** that defines the morphology null. C_i — and therefore `f_max` — is reported **separately for each family**, as a function of (depth, period): a flat megastructure occulter is easy and bounds tightly, a subtle tilted shape at shallow depth is hard and bounds loosely, and the limit *states which is which* rather than collapsing every anomaly onto one box (which would silently overstate completeness for exactly the non-box signals this channel exists to find). A synthetic-only pilot (`runners/pilot_injection.py`, run pre-freeze, no target data) confirms the morphology metrics (flat-bottom, ingress/egress asymmetry, depth-CV) separate these families from the planet locus above explicit *folded-profile per-bin SNR* floors — at 1 % depth, box $\gtrsim$ 20 on flat-bottom, tail $\gtrsim$ 12 on asymmetry (plus depth-CV ≈ 0.7 vs ≈ 0 for a stable transiter), while subtle tilted/triangular shapes are *not* separable at shallow depth and need depth-CV or deeper transits. The metrics demonstrably fire on the non-box signals, and the SNR floors are reported, not hidden.
- **Stopping rule (unchanged from Phase 1):** "unexplained" means "survives the registered battery at procedure-frozen thresholds," nothing more. We report the *full ranked residual list*, and the standing interpretation is that a residual is a target for conventional astrophysical follow-up — never, in this document or its outputs, a claim of detected intelligence.

6. Analysis plan

1. Freeze the sample → the immutable checksummed manifest (§3).
2. Pull TESS/Kepler light curves (deterministic recipe); detrend.
3. **Calibration & validation — a pre-registered *procedure*, not a frozen number.** We do not peek at data to pick thresholds, and we do not register guessed values. We register the *method* below in full; the resulting numbers (the per-cohort nulls, the genomic-control λ, the FWER σ-bar, the per-family C_i) are *outputs* of running it,

generated after registration and auditable by a reviewer against this specification. Every sub-step touches only the noise floor and synthetic injections — **never the real candidate tail** — so executing it reveals no candidate. - **3a. Outlier-blind noise floor & cohorts.** For every manifest star, compute the noise metric as the **MAD of the out-of-transit continuum after iterative downward 3σ-clipping** (dips masked, §5(a)); assign each star to a noise/activity cohort by **fixed percentile edges** of that MAD (and the activity proxies of §3), registered here so the binning is reproducible. Acceptance check: an injected deep dip must *not* move its star's cohort (the clip prevents an anomaly inflating its own baseline). - **3b. Bulk sensitivity (U-shaped transits).** Inject **constant-depth, limb-darkened (U-shaped) transits** across a registered depth × period grid into the real light curves; run BLS/TLS recovery. This yields the per-star, per-cohort completeness C_i for ordinary transits, the genomic-control inflation λ, and the **single FWER σ-bar** set so the expected pure-noise false alarms over the whole manifest N_{total} is < 1 (§5(b)). Whatever σ this procedure returns is the bar — we expect ≈ 6–7σ, but we report the value it yields. - **3c. Anomaly sensitivity & metric validation (the frozen library, §5).** Inject the registered anomaly families — **variable-depth disintegrating tails and strictly asymmetric dips** — across the same grid, and confirm (i) the **variable-depth / aperiodic detector recovers them** (BLS does not silently discard a fluctuating-depth dip as noise) and (ii) the **morphology metrics flag them non-planetary**. This yields the **per-family C_i** for the per-family `f_max`. Recall (§4) that detection does not depend on morphology: 3b sets detection sensitivity, 3c verifies the parallel nets catch the non-planetary minority. - **3d. Known-object controls.** Validate the end-to-end detector + metrics against **named, published systems** (not the target tail): they must recover ordinary transiters and must fire on real anomalous transiters — KIC 12557548 / Kepler-1520 (disintegrating dust tail) and WD 1145+017 — reproducing their known behaviour. This is the Phase-1 "validate on WD 1856+534 b" discipline, registered as a step rather than left informal.

All outputs of step 3 are frozen before step 4; none is ever re-touched in reaction to a candidate. 4. **Unblind:** run the BLS/TLS + variable-depth search, apply the frozen threshold, compute morphology metrics, and pass survivors through the natural-explanation battery (automated centroid gate first). 5. Report the full ranked residual list and the calibrated per-family `f_max(depth, period)`.

Same recipe-in-repo, checksum, gitignored-bulk discipline as Phase 1. The split between step 3 (noise-floor/synthetic calibration + known-object validation, all outputs frozen) and step 4 (real candidates unblinded against the frozen thresholds) is the integrity crux at this sample size: it is what prevents a strange real signal from tempting a threshold tweak. A reviewer judges us not on the numbers we obtained but on whether we executed the registered procedure that produced them.

7. Outcomes and interpretation

- **Expected:** a clean, explained null plus a prevalence upper limit, with a byproduct catalogue of K-dwarf transit candidates and activity characterisation of value independent of the technosignature framing.

- **A surviving residual** is, in order of probability, a *previously-unmodelled natural* occulter or variability class — published as such, with all natural tests shown. A *fluctuating or structured* anomalous morphology would be the single highest-value outcome. No claim of artificiality is made on the basis of this pipeline.

8. Open science and reproducibility

Public repository; pinned data recipe, query text, frozen manifests, and checksums; bulk data fetched on demand (gitignored). The analysis runs on the **validated population-agnostic core** plus a new **K-dwarf plugin** — Phase 1 reused, not forked, with the WD regression test as a standing guarantee that the shared core is unchanged. AI-collaboration transcripts archived. Registered on OSF before any K-dwarf light curve is analysed; post-registration changes are dated, public amendments.

Statement of Provenance and Acknowledgments

Human accountability. Authored and directed by the sole investigator (T. Loewald), who bears full responsibility for its contents, methodological selections, and errors. Special thanks are owed to a cross-disciplinary collaborator whose insight — bringing the population-level statistical standards of epidemiology and genome-wide association studies to bear on an astrophysical problem — motivated the empirical-null and genomic-control framework this search inherits from Phase 1.

AI intellectual provenance. Two AI systems (Google Gemini, Anthropic Claude) functioned as active computational logic engines and co-designers in methodology, implementation, and adversarial review; they are credited as such while the investigator retains all accountability. The complete, lightly-redacted working transcripts are archived in the public repository.

References

(Shared methodology references as Phase 1 — Efron 2004 empirical null; Devlin & Roeder 1999 genomic control; Benjamini & Hochberg 1995 / Storey 2002 FDR; Gross & Vitells 2010 look-elsewhere; Jenkins et al. 2002 Kepler 7.1σ — plus transit-specific below.) - Arnold, L. F. A. 2005, *ApJ*, 627, 534 — artificial/ringed transit signatures. - Kovács, G., Zucker, S., & Mazeh, T. 2002, *A&A*, 391, 369 — Box Least Squares. - Hippke, M., & Heller, R. 2019, *A&A*, 623, A39 — Transit Least Squares. - Boyajian, T. S., et al. 2016, *MNRAS*, 457, 3988 — KIC 8462852 (anomalous, structured transits). - Rappaport, S., et al. 2012, *ApJ*, 752, 1 — KIC 12557548, a disintegrating planet (step-3d control). - Vanderburg, A., et al. 2015, *Nature*, 526, 546 — disintegrating planetesimals at WD 1145+017. - Angus, R., et al. 2019, *AJ*, 158, 173 — gyrochronology age–rotation relation (youth floor, §3). - Claret, A. 2017, *A&A*, 600, A30 — limb-darkening coefficients (TESS), Channel-B forward models. - Borucki, W. J., et al. 2010, *Science*, 327, 977 — Kepler. - Ricker, G. R., et al. 2015, *JATIS*, 1, 014003 — TESS. - Jenkins, J. M., Caldwell, D. A., & Borucki, W. J. 2002, *ApJ*, 564, 495 — transit detection threshold. - Efron, B. 2004, *JASA*, 99, 96 — empirical null. · Devlin & Roeder 1999, *Biometrics*, 55, 997 — genomic control. - Gentile Fusillo, N. P., et al. 2021, *MNRAS*, 508, 3877 / Gaia Collaboration — sample provenance.